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BY

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PHYSICS 2023

(iii) X-rays of wavelength $2.64 \times 10^{-10} m$ are used in Compton scattering process. Find the fractional change in wavelength for a scattering angle of 120^o.
(Give = $6.63 \times 10^{-34} Js, m_o = 9.1 \times 10^{-31} kg, c = 3 \times 10^8$)

DATA:

$\lambda_1 = 3.64 \times 10^{-10}$

$\theta = 120^o$

$h = 6.63 \times 10^{-34} Js$

$m_o = 9.1 \times 10^{-31} kg$

$c = 3 \times 10^8 m/s$

fractional change $= \frac{\Delta\lambda}{\lambda_1} = ?$

SOLUTION:

$$\because \Delta\lambda = \frac{h}{m_o c}(1 - \cos\theta)$$

Divided by λ_1 on both sides

$$\frac{\Delta\lambda}{\lambda_1} = \frac{h}{m_o c \lambda_1}(1 - \cos\theta)$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{6.63 \times 10^{-34}}{(9.1 \times 10^{-31})(3 \times 10^8)(3.64 \times 10^{-10})}(1 - \cos 120)$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{6.63 \times 10^{-34}}{9.1 \times 3 \times 3.64 \times 10^{-31+8-10}}(1 + 0.5)$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{9.945 \times 10^{-34}}{9.9372 \times 10^{-32}}$$

$$\frac{\Delta\lambda}{\lambda_1} = 1.0 \times 10^{-2}$$

$$\frac{\Delta\lambda}{\lambda_1} = 0.010$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{0.010}{100}$$

$$\frac{\Delta\lambda}{\lambda_1} = 1\% \textit{ Answer}$$

(v) The half-life of ${}_{104}Po^{210}$ is 140 days. By what percent does its activity will decrease per week?

DATA:

$T_{\frac{1}{2}} = 140\, day = \frac{140}{7} = 20\, weeks$

$\lambda = ?$

SOLUTION:

$$\because T_{\frac{1}{2}}\lambda = 0.693$$

$$\lambda = \frac{0.693}{T_{\frac{1}{2}}}$$

$$\lambda = \frac{0.693}{20}$$

$$\lambda = 0.03465$$

FOR PERCENTAGE

$$\lambda = 0.03465 \times 100$$

$$\lambda\, in\, percent = 3.465\,\%$$

(vi) Calculate the temperature at which the root mean square speed of nitrogen molecule is $3300ms^{-1}$.

DATA:

$v_{rms} = 3300\, m/s$

$m = \frac{M}{N_A} = \frac{28g}{6.02 \times 10^{23}} = 4.65 \times 10^{-23} g = 4.651 \times 10^{-26} kg$

$K = 1.38 \times 10^{-23} J/K$

SOLUTION:

$$\because v_{rms} = \sqrt{\frac{3KT}{m}}$$

Squaring Both Sides

$$v_{rms}{}^2 = \frac{3KT}{m}$$

$$\frac{v_{rms}{}^2 m}{3K} = T$$

$$T = \frac{(3300)^2(4.651 \times 10^{-26})}{3(1.38 \times 10^{-23})}$$

$$T = \frac{(1.089 \times 10^7)(4.651 \times 10^{-26})}{4.14 \times 10^{-23}}$$

$$T = \frac{5.0 \times 10^{-19}}{4.14 \times 10^{-23}}$$

$$T = 12234.15\, K$$

$$T = 12234.15 - 273$$

$$T = 12234.15 - 273$$

$$T = 11961.58°C$$

(ix) A particle of charge $3.2 \times 10^{-19} C$ and mass $2.48 \times 10^{-27}\, kg$ is held motionless between two horizontal parallel plates separated by $5cm$. Find the potential difference between the plates.

DATA:

$m = 2.48 \times 10^{-27}\, kg$
$q = 3.2 \times 10^{-19} C$
$d = 5\, cm = 0.05m$
$\Delta V = ?$

SOLUTION:

$$\because \Delta V = E \times d - - - (1)$$

In this case particle between two horizontal parallel plates is motionless, so

$$F_{Elec.} = W_{particle}$$
$$Eq = mg$$
$$E = \frac{mg}{q}$$
$$Ee = \frac{(2.48 \times 10^{-27})(9.8)}{3.2 \times 10^{-19}}$$
$$E = 7.595 \times 10^{-8}\, Volt/meter$$

Substituting value of "E" in Equation (1)

$$\Delta V = 7.595 \times 10^{-8} \times 0.05$$
$$\Delta V = 3.7975 \times 10^{-9} volt \quad Answer$$

(xii) Find the speed at which mass of particle will be doubled.

DATA:

$m = 2m_o$
$c = 3 \times 10^8 m/s$
$V = ?$

$$\because m = \frac{m_o}{\sqrt{1 - \frac{V^2}{c^2}}}$$
$$2\cancel{m_o} = \frac{\cancel{m_o}}{\sqrt{1 - \frac{V^2}{c^2}}}$$
$$2 = \frac{1}{\sqrt{1 - \frac{V^2}{c^2}}}$$
$$2\sqrt{1 - \frac{V^2}{c^2}} = 1$$
$$\sqrt{1 - \frac{V^2}{c^2}} = \frac{1}{2}$$

Squaring on both sides

$$1 - \frac{V^2}{c^2} = \frac{1}{4}$$
$$1 - \frac{1}{4} = \frac{V^2}{c^2}$$

$$\frac{4-1}{4}=\frac{V^2}{c^2}$$
$$V^2=\frac{3}{4}c^2$$
$$V=\frac{\sqrt{3}}{2}c$$

OR

$$V=\frac{\sqrt{3}}{2}(3\times10^8 m/s)$$
$$V=259807621. m/s$$

(xiii) Three resistances each of 6Ω can be connected in four different ways. What is equivalent resistance of each combination?

DATA:

$R_1=R_2=R_3=6\Omega$

$R_e=?$

SOLUTION:

1[st] way:

Connected in series

$$\because R_e=R_1+R_2+R_3$$
$$R_e=6+6+6$$
$$R_{e1}=18\Omega$$

2[nd] way:

Connected in parallel

$$\because\frac{1}{R_{e2}}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}$$
$$\frac{1}{R_{e2}}=\frac{1}{6}+\frac{1}{6}+\frac{1}{6}$$
$$\frac{1}{R_{e2}}=\frac{3}{6}$$
$$R_{e2}=\frac{6}{3}=2\Omega$$

3[rd] way:

Connected two in series and one in parallel

$$R_e{}'=R_1+R_2$$
$$R_e'=6+6$$
$$R_e'=12\Omega$$
$$\frac{1}{R_{e3}}=\frac{1}{R_e'}+\frac{1}{R_3}$$
$$\frac{1}{R_{e3}}=\frac{1}{12}+\frac{1}{6}$$
$$\frac{1}{R_{e3}}=\frac{1+2}{12}$$
$$R_{e3}=\frac{12}{3}$$
$$R_{e3}=4\Omega$$

4[th] way:

Connected two in parallel and one in series

$$\because\frac{1}{R_e{}''}=\frac{1}{R_1}+\frac{1}{R_2}$$
$$\frac{1}{R_e{}''}=\frac{1}{6}+\frac{1}{6}$$
$$\frac{1}{R_e{}''}=\frac{2}{6}$$
$$R_e''=\frac{6}{2}$$
$$R_e''=3\Omega$$
$$\because R_{e4}=R_e''+R_3$$
$$R_{e4}=3+6$$
$$R_{e4}=9\,\Omega\ \ \textit{Answer}$$

(xiv) An iron core solenoid with 400 turns has a cross-section area of $4.0cm^2$. A current of 2.0 ampere passing through it produces $B = 0.2\,weber/m^2$. What emf is produced in it if the current is turned off in $0.1\,sec$? What is its self-inductance?

DATA:

$N = 400$

$A = 2.0\text{ cm}^2 = \dfrac{4}{100 \times 100} = 4 \times 10^{-4} m^2$

$\Delta I = (2.0 - 0) = 2.0\,Ampere$

$B = 0.2\text{ weber/m}^2$

$e.m.f. = \xi =$

$\Delta t = 0.1\,second$

Self Inductance $= L = ?$

SOLUTION:

$$\because \xi = N.\frac{\Delta\phi}{\Delta t}$$

$$But\ \Delta\phi = B.\Delta A$$

$$\therefore \xi = N.\frac{B.\Delta A}{\Delta t}$$

$$\xi = (400)\frac{(0.2)(4 \times 10^{-4})}{(0.1)}$$

$$\xi = \frac{320}{0.1} \times 10^{-4}$$

$$\xi = 0.32\,Volts$$

$$\because \xi = L\frac{\Delta I}{\Delta t}$$

$$L = \frac{\xi\,\Delta t}{\Delta I}$$

$$L = \frac{(0.32)(0.1)}{(2.0)}$$

$$L = \frac{0.032}{2.0}$$

$$L = 0.016\,Henry$$

$$L = 0.016 \times 1000$$

$$L = 16\,mH\ \ Answer$$

PHYSICS 2022

2. i) A Carnot engine works between 800°C and 400°C, if source temperature is increased by 50°C or sink temperature is decreased by 50°C, then which will cause greater efficiency.

DATA:

$T_1 = 800°C$

$T_2 = 400°C$

$T_1' = 800 + 50 = 850$

$T_2' = 400°C - 50 = 350K$

SOLUTION:

The efficiency of the cycle when source temperature is increased by 50°C.

The efficiency of the cycle when source temperature is increased by 50°C.

$$\because E_1 = \left(1 - \frac{T_2}{T_1'}\right) \times 100$$

The efficiency of the cycle when source temperature is decreased by 50°C.

$$\because E_2 = \left(1 - \frac{T_2'}{T_1}\right) \times 100$$

$$E_1 = \left(1 - \frac{400}{850}\right) \times 100$$
$$E_1 = (1 - 0.47) \times 100$$
$$E_1 = 0.53 \times 100$$
$$E_1 = 53$$

$$E_2 = \left(1 - \frac{350}{800}\right) \times 100$$
$$E_2 = (1 - 0.4375) \times 100$$
$$E_2 = 0.5625 \times 100$$
$$E_2 = 56.25$$

Hence $E_2 > E_1$ and reducing the temperature of the sink by 50°C improves the efficiency of heat engine

ii) "What amount of current is required to produce a magnetic field of 37.2 Tesla at a distance of 2 cm from a long straight current carrying wire? ($\mu_o = 4\pi \times 10^{-7}$ Weber/Am)

DATA:

$B = 37.2\ Tesla$

$r = 2\ cm = 0.02m$

$\mu_o = 4\pi \times 10^{-7}$ Weber/Am

$I = ?$

SOLUTION:

The expression of magnetic field due to straight current carrying conductor can be expressed as follows.

$$\because B = \frac{\mu_o I}{2\pi r}$$
$$I = \frac{B \times 2\pi r}{\mu_o}$$
$$I = \frac{(37.2) \times 2(\pi)(0.02)}{(4\pi \times 10^{-7})}$$
$$I = \frac{(37.2) \times 2(0.02)}{(4 \times 10^{-7})} = \frac{1.488}{4} \times 10^7$$
$$I = 0.37 \times 10^7 = 3.7 \times 10^6\ Amp$$

iii) In photo electric experiment, a stopping potential of 1.25 volt is needed. Find the maximum kinetic energy and maximum speed of photo electron $= 1.6 \times 10^{-19} C$; $m_e = 9.1 \times 10^{-31}\ kg$

DATA:

$V = 1.25\ volt$

$e = 1.6 \times 10^{-19} C$

$m_e = 9.1 \times 10^{-31}\ kg$

$K.E_{max} = ?$

$v_{max} = ?$

SOLUTION:

Kinetic energy gained by an electron when accelerated by Potential Diff "V" is:

$$\because K.E_{max} = V \times e$$
$$K.E_{max} = 1.25 \times 1.6 \times 10^{-19}$$
$$K.E_{max} = 1.25 \times 1.6 \times 10^{-19}$$
$$K.E_{max} = 2 \times 10^{-19} J$$

Also

$$\frac{1}{2}m_e v^2 = 2 \times 10^{-19} J$$

$$v^2 = \frac{2 \times 2 \times 10^{-19} J}{m_e}$$

$$v^2 = \frac{4 \times 10^{-19} kg\, m^2 s^{-2}}{9.1 \times 10^{-31} kg}$$

$$v^2 = \frac{4}{9.1} \times 10^{-19+31} m^2 s^{-2}$$

$$v^2 = \frac{4}{9.1} \times 10^{-19+31} m^2 s^{-2}$$

$$v^2 = 4.3956 \times 10^{11} m^2 s^{-2}$$

$$v = 662993.54\ ms^{-1}$$

vii) A charge particle of charge 2×10^{-9}C, in an electric field between two parallel metal plates 4 cm apart, is acted upon by a force of 10^{-4}N

a) What is the intensity of the electric field?

b) What is potential difference between the plates,

DATA:

$q = 2 \times 10^{-9} C$

$d = 4\ cm = 0.04 m$

$F = 10^{-4} N$

$E = ?$

$\Delta V = ?$

SOLUTION:

$$\because F = q \times E$$

$$E = \frac{F}{q} = \frac{10^{-4}}{2 \times 10^{-9}} = \frac{1}{2} \times 10^{-4+9}$$

$$E = \frac{10^5}{2} = \frac{100000}{2}$$

$$E = 50000\ \text{v/m}$$

$$\Delta V = E \times d$$

$$\Delta V = 50000 \times 0.04$$

$$\Delta V = 2000\ volt$$

ix) A 400 volt voltmeter has a total resistance of 40,000 Ω. What additional series resistance mug be connected to it to increase the range to 750 volts?

DATA:

$V_1 = 400\ volts$

$R_v = 40,000\Omega$

$V_2 = 750\ volts$

$R_x =$

SOLUTION:

$$V_1 = IR_v$$

$$I = \frac{V_1}{R_x}$$

$$\because R_x = \frac{V_2}{I} - R_v$$

$$R_x = \frac{750}{0.1} - 40000$$

$$I = \frac{400}{40000} = 0.1\,Ampere$$

$$R_x = 75000 - 40000$$
$$R_x = 35000\Omega \quad Answer$$

ii) Hydrogen atom in ground state is excited by absorbing a photon of 12.15eV. Find the quantum number of this excited state.

DATA:

$E_1 = -13.6\,eV$

$E_2 = 12.15\,eV$

$n = ?$

SOLUTION:

$$\because E_n = E_1 + E_2$$
$$E_n = -13.6 + 12.15$$
$$E_n = -1.45\,eV$$

Energy of electron in any state is given by

$$E_n = \frac{1}{n^2}(-13.6)\,eV$$

$$-1.45\,eV = \frac{1}{n^2}(-13.6)eV$$
$$n^2 = \frac{13.6}{1.45}$$
$$n^2 = 9.38$$
$$n = 3.06$$

Quantum number of the excited state is 3

iii) If the number of atoms per gram of $_{88}Ra^{226}$ is 2.666×10^{21} and it decays with the half life of 1622 year. Find decay constant and activity of the sample.

DATA:

$^{226}_{88}Ra = N = 2.666 \times 10^{21}\,atoms$

$T_{1/2} = 1622 \times 365 \times 24 \times 60 \times 60 = 5.115 \times 10^{10}\,seconds$

$\lambda = ?$

$R = ?$

SOLUTION:

$$T_{1/2} = \frac{0.693}{\lambda}$$
$$\lambda = \frac{0.693}{T_{1/2}}$$
$$\lambda = \frac{0.693}{5.115\times10^{10}}$$
$$\lambda = 1.345 \times 10^{-11}\,per\,Sec$$

$$\frac{\Delta N}{\Delta t} = \lambda N$$
$$But\ R = \frac{\Delta N}{\Delta t}$$
$$R = (1.345 \times 10^{-11})(2.666 \times 10^{21})$$
$$R = 3.5859 \times 10^{10}$$

$$R = 3.5859 \times 10^{10}\ \text{Disintegration / sec. } Answer$$

PHYSICS 2021

ii) How many electrons per second pass through the cross section of a wire carrying a current of 0.7 ampere?

DATA:

Current = $I = 0.7$ ampere

Time = $t = 1$ second

Charge on Electron = $Q = 1.6 \times 10^{-19}C$

No. of Electrons = $n = ?$

SOLUTION:

$$\because I = \frac{Q}{t}$$

And

$$Q = ne$$

$$\therefore I = \frac{ne}{t}$$

$$n = \frac{I \times t}{e}$$

$$n = \frac{0.7[illegible]1}{1.6 \times 10^{-19}}$$

$$n = \frac{6.7}{1.6} \times 10^{19}$$

$$n = 4.357 \times 10^{18}$$

Hence, the number of electrons per second passing through the wire are -4.357×10^{18}

iii) A galvanometer has a resistance of 50Ω and it deflects full scale when a current of 500 micro ampere flows in it. How can it be converted into ammeter of range 15 ampere and voltmeter of range 300 volts?

DATA:

$R_g = 50$ ohm

$I_g = 500\mu = \frac{500}{10^6} = 5 \times 10^{-4}A$

Range of ammeter $= I = 15$

Range of voltmeter $= V = 300$

Shunt Resistance $= R_s = ?$

Series Resistance $= R_x = ?$

SOLUTION:

$$\because R_s = \frac{I_g R_g}{(I - I_g)}$$

$$R_s = \frac{(5 \times 10^{-4})(50)}{(15 - 5 \times 10^{-4})}$$

$$R_s = \frac{0.025}{14.9995}$$

$$R_s = 0.00166 \text{ ohm}$$

$$\because R_x = \frac{V}{I_g} - R_g$$

$$R_x = \frac{300}{5 \times 10^{-4}} - 50$$

$$R_x = 600000 - 50$$

$$R_x = 5.99995 \text{ ohms}$$

v) Light of wavelength 486.3nm is emitted by a hydrogen atom in Balmer series. What transition of hydrogen atom is responsible for this radiation? ($R_H = 1.097 \times 10^7\ m^{-1}$)

DATA:

Initial Istat $= n_f = 2$

wavelength $= \lambda = 486.3\text{nm} = 486.3 \times 10^{-9}\text{m}$

Initial Istat $= n_i = ?$

Rydberg constat $= R_H = 1.097 \times 10^7 m^{-1}$

SOLUTION:

$$\because \frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{486.3 \times 10^{-9}} = (1.097 \times 10^7)\left(\frac{1}{(2)^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{(486.3 \times 10^{-9})(1.097 \times 10^7)} = \left(\frac{1}{4} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{(486.3\times 1.097)\times 10^{-9+7}}=\left(0.25-\frac{1}{n_i^2}\right)$$

$$\frac{1}{5.335}=\left(0.25-\frac{1}{n_i^2}\right)$$

$$0.1875=0.25-\frac{1}{n_i^2}$$

$$\frac{1}{n_i^2}=\frac{1}{16}$$

$$n_i^2=16$$

$$n_i=4$$

$$\frac{1}{\lambda}=(1.097\times 10^7)\left(\frac{1}{(2)^2}-\frac{1}{(\infty)^2}\right)$$

$$\frac{1}{\lambda}=1.097\times 10^7\left(\frac{1}{4}-\frac{1}{\infty}\right)$$

$$\frac{1}{\lambda}=1.097\times 10^7\left(\frac{1}{4}-0\right)$$

$$\frac{1}{\lambda}=1.097\times 10^7\left(\frac{1}{4}\right)$$

$$\frac{1}{\lambda}=2.742\times 10^6$$

$$\lambda=\frac{1}{2.742\times 10^6}$$

$$\lambda=3.647\times 10^{-7}\,Meter$$

vi) Sodium surface is shined with light of wavelength 3×10^{-7} **m. If the work function of Na is 2.46 eV. Find the K.E of the photoelectrons and cut off wavelength** λ_c

$h=6.63\times 10^{-34}$ J.s, $c=3\times 10^8$ m/s, 1eV $=1.6\times 10^{-19}$ J.

DATA:

$\lambda=3\times 10^{-7}m$

$\phi=2.46\,ev=2.46\times 1.6\times 10^{-19}$

$=3.936\times 10^{-19}J$

$h=6.63\times 10^{-34}\,J.s$

$c=3\times 10^8 m/s$

$K.E.=?$

$\lambda_c=?$

SOLUTION:

$$\frac{hc}{\lambda}=\phi+K.E_{max}$$

$$\frac{(6.63\times 10^{-34})(3\times 10^8)}{(3\times 10^{-7})}=3.936\times 10^{-19}+K.E_{max}$$

$$K.E_{max}=\frac{19.89}{3}\times 10^{-34+8+7}-3.936\times 10^{-19}$$

$$K.E_{max} = 6.63 \times 10^{-19} - 3.936 \times 10^{-19}$$

$$K.E_{max} = 2.694 \times 10^{-19} J$$

$$K.E_{max} = \frac{2.694 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$K.E_{max} = 1.682\ eV$$

$$\because \phi = \frac{hc}{\lambda_c}$$

$$\lambda_c = \frac{hc}{\phi} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{3.936 \times 10^{-19}}$$

$$= \frac{19.89 \times 10^{-26}}{3.936 \times 10^{-19}}$$

$$\lambda_c = 5.053 \times 10^{-7} m$$

$$\lambda_c = 5.053 \times 10^{-7} \times 10^{10}$$

$$\lambda_c = 5.053 \times 10^3\ \mathring{A}$$

viii) A proton of mass 1.67×10^{-27} kg and charge 1.6 x 10-19 C is to be held motionless between two horizontal parallel plates. The potential difference applied between plates $is\ 1.02 \times 10^{-8}$ volts. Calculate the distance between plates.

DATA:

$Mass\ of\ Proton = m = 1.67 \times 10^{-27} kg$

$Charge\ on\ Proton = q = 1.6 \times 10^{-19} C$

$potential\ difference = \Delta V = 1.02 \times 10^{-8} Volts$

$distance = d = ?$

SOLUTION:

$$\because \Delta V = E \times d$$

$$d = \frac{\Delta V}{E} - - - - (1)$$

In this case particle between two horizontal parallel plates is motionless, so

$$\because F_{Elec} = W_{particle}$$

$$Eq = mg$$

$$E = \frac{mg}{q}$$

$$Ee = \frac{(1.67 \times 10^{-27})(9.8)}{1.6 \times 10^{-19}}$$

$$E = 1.022 \times 10^{-26+19}$$

$$E = 1.022 \times 10^{-7}\ Volt/meter$$

Substituting value of "E" in Equation (1)

$$d = \frac{1.02 \times 10^{-8}}{1.022 \times 10^{-7}}$$

$$d = \frac{1.02}{1.022} \times 10^{-8+7}$$

d = 0.0998 meter Answer

PHYSICS 2019

2. (ii) If the number of atoms per gram of $_{88}Ra^{266}$ is 2.666×10^{21} and it decays with the half-life of 1622 years, find the decay constant and activity of the sample. ($1\ Year = 3.15 \times 10^{21}$)

DATA:

$^{226}_{88}Ra = N = 2.666 \times 10^{21} atoms$

$T_{1/2} = 1622 \times 365 \times 24 \times 60 \times 60 = 5.115 \times 10^{10} seconds$

$\lambda = ?$

$R = ?$

SOLUTION:

$T_{1/2} = \frac{0.693}{\lambda}$

$\lambda = \frac{0.693}{T_{1/2}}$

$\lambda = \frac{0.693}{5.115 \times 10^{10}}$

$\lambda = 1.345 \times 10^{-11}$ per Sec

$\frac{\Delta N}{\Delta t} = \lambda N$

But $R = \frac{\Delta N}{\Delta t}$

$R = (1.345 \times 10^{-11})(2.666 \times 10^{21})$

$R = 3.5859 \times 10^{10}$

$R = 3.5859 \times 10^{10}$ Disintegration / sec. *Answer*

2. (iii) A coil of 400 turns in AC Generator having an area of 0.1 m^2 is rotating in a magnetic field of 50 T. in order to generate a maximum voltage of 220 volts, how fast is the coil to be rotated? Express your answer in revolutions/second.

DATA:

N = 400 turns

$A = 0.1 m^2$

B = 50 T

$E_{max} = 220$ Volts

$\omega = ?$

SOLUTION:

$\because E_{max} = BNA\omega$

$\omega = \frac{V_{max}}{BNA}$

$\omega = \frac{220}{(50)(400)(0.1)}$

$\omega = \frac{220}{2000}$

$= 0.11\ rad/sec$ Rev. per Sec

$\omega = \frac{0.11}{2\pi} = \frac{0.11}{6.284}$

$\omega = 0.0175$ rev./sec.

$N = \frac{E_{max}}{BA\omega}$

$N = \frac{149}{(0.06)(0.05)(79)}$

$N = \frac{149}{0.237}$

$N = 628\ turns$

2. (vi) What is the wavelength of 3rd spectral line of Paschen Series in hydrogen atom? ($R_H = 1.097 \times 10^7 m^{-1}$)

DATA:

$n_i = 6$

$n_f = 3$

$R_H = 1.097 \times 10^7\ m^{-1}$

$\lambda = ?$

SOLUTION:

FOR 3rd spectral line in Paschen Series

$$\because \frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda} = (1.097 \times 10^7)\left(\frac{1}{(3)^2} - \frac{1}{(6)^2}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{1}{9} - \frac{1}{36}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{4-1}{36}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{3}{36}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{1}{12}\right)$$

$$12 = 1.097 \times 10^7 \lambda$$

$$\frac{12}{1.097 \times 10^7} = \lambda$$

$$\lambda = 10.938 \times 10^{-7}\, m \quad Answer$$

2. (viii) A rectangular bar of iron is 2cm × 2cm in cross section and 20cm long. What will be its resistance at 500°C?

DATA:

$A = 2cm \times 2cm = 4cm = \frac{4}{100} = 0.04m^2$

$L = 20cm = \frac{20}{100} = 0.2m$

$t = 500°C = 500 + 273 = 773K$

$\alpha = 0.0052k^{-1}$

$\rho = 11 \times 10^{-8}\Omega m$

$R_t = R_{773} = ?$

SOLUTION:

$$\because R_t = R_o(1 + \alpha t) --- (1)$$

$$R_o = \rho\frac{L}{A}$$

$$R_o = \frac{(11 \times 10^{-8})(0.2)}{0.04}$$

$$R_o = \frac{2.22}{0.04} \times 10^{-7}$$

$$R_o = 5.55 \times 10^{-6}\Omega$$

Now

$$R_{773} = (5.55 \times 10^{-6})[1 + (0.0052)(773)]$$

$$R_{773} = (5.55 \times 10^{-6})(1 + 4.0196)$$

$$R_{773} = (5.55 \times 10^{-6})(5.0196)$$

$$R_{773} = 2.230 \times 10^{-5}\Omega \quad Answer$$

2. (ix) Two capacitors of $2\mu F$ and $4\mu F$ are connected in series to 40 volt Battery. Calculate the charge on these capacitors and potential difference across each.

DATA:

$C_1 = 2\ \mu F = 2 \times 10^{-6}F$

$C_2 = 4\ \mu F = 4 \times 10^{-6}F$

$V = 40\ \mu F$

$Q_1 = ?$

$Q_2 = ?$

$V_1 = ?$

SOLUTION:

Since both capacitors are connected in series therefore

$$\frac{1}{C_e}=\frac{1}{C_1}+\frac{1}{C_2}$$

$$\frac{1}{C}=\frac{1}{2}+\frac{1}{4}$$

$$\frac{1}{C_e}=\frac{2+1}{4}=\frac{3}{4}$$

$$C_e=\frac{4}{3}$$

$$C_e=1.33\ \mu F$$

OR

$$C_e=1.33\times10^{-6}F$$

Total Charge stored in both capacitors is

$$\because Q=C_eV$$

$$Q=(1.33\times10^{-6}F)(40)$$

$$Q=5.33\times10^{-5}Coulomb$$

Since both capacitors are connected in series therefore charge on each capacitor will be

$$Q=Q_1=Q_2=5.33\times10^{-5}Coulomb$$

POTENTIAL DIFFERENCE ACROSS 1ST CAPACITOR

$$\because Q_1=C_1V_1$$

$$V_1=\frac{Q_1}{C_1}=\frac{5.33\times10^{-5}}{2\times10^{-6}F}$$

$$V_1=26.65\ Volts$$

POTENTIAL DIFFERENCE ACROSS 2ND CAPACITOR

$$\because Q_2=C_2V_2$$

$$V_2=\frac{Q_2}{C_2}=\frac{5.33\times10^{-5}}{4\times10^{-6}F}$$

$$V_2=13.325\ Volts$$

2. (x) Find the change in volume of an aluminum sphere of 0.4m radius when it is heated from 0°C to 100°C. ($\propto 24\times10^{-6}C^{-1}$)

DATA:

$r=0.4m$

$T_1=0°C$

$T_2=100°C$

$\Delta T=T_2-T_1=100-0=100°C$

$\propto=24\times10^{-6}C^{-1}$

$\beta=3(24\times10^{-6})C^{-1}=7.2\times10^{-5}C^{-1}$

$\Delta V=?$

$$\because V=\frac{4}{3}\pi r^3$$

$$V_1=\frac{4}{3}\pi(0.4)^3$$

$$V_1=(1.333)(3.142)(0.064)$$

$$V_1=0.268\ m^3$$

For Initial volume of sphere

For change in volume of

$$\because \Delta V=V_1\beta\Delta T$$

$$\because \Delta V=(0.268)(7.2\times10^{-5})(100)$$

$$\because \Delta V=1.93\times10^{-3}m^3$$

$$\because \Delta V=0.00193m^3$$

2. (xv) What will be the velocity and momentum of a particle whose rest mass is m_o and kinetic energy is equal to twice of its rest mass energy

DATA:

$m_o=E$

$K.E=2\,E_o$

$V=?$

$P=?$

$$Total\ Energy = Rest\ mass\ energy + K.E$$
$$Total\ Energy = E_o + 2E_o$$
$$E = 3E_o$$
$$But\quad E_o = m_oc^2 \quad and\ E = mc^2$$
$$\therefore mc^2 = 3m_oc^2$$
$$m = 3m_o$$
$$\because m = \frac{m_o}{\sqrt{1-\frac{V^2}{C^2}}}$$
$$m = \frac{m_o}{\sqrt{1-\frac{V^2}{C^2}}}$$
$$3m_o = \frac{m_o}{\sqrt{1-\frac{V^2}{C^2}}}$$
$$3 = \frac{1}{\sqrt{1-\frac{V^2}{C^2}}}$$

Squaring both sides
$$9 = \frac{1}{1-\frac{V^2}{C^2}}$$
$$1-\frac{V^2}{C^2} = \frac{1}{9}$$
$$1-\frac{1}{9} = \frac{V^2}{C^2}$$
$$\frac{9-1}{9} = \frac{V^2}{C^2}$$
$$\frac{V^2}{C^2} = \frac{8}{9}$$
$$V^2 = \frac{8}{9}C^2$$
$$V = \sqrt{\frac{8}{9}C^2}$$
$$V = \frac{2}{3}\sqrt{2}\,c\ Answer$$

$$\because P = mV$$
$$P = (3m_o)\left(\frac{2}{3}\sqrt{2}\,c\right)$$
$$P = 2\sqrt{2}\,m_oc\ \ Answer$$

PHYSICS 2018

(ii) Find the potential difference across the two ends of 15m long copper wire 0.5 mm in diameter to maintain steady current of 4 amperes. (Resistivity of copper= $1.54 \times 10^{-8}\Omega m$).

DATA:

$L = 15\ m$

$D = 0.5mm = \frac{0.5}{1000} = 5 \times 10^{-4}m$

$$r = \frac{5 \times 10^{-4}}{2}m = 2.5 \times 10^{-4}m$$

$I = 4\ Amperes$

$\rho = 1.54 \times 10^{-8}\Omega m$

$V = ?$

SOLUTION:

$$\because A = \pi r^2$$
$$A = \left(\frac{22}{7}\right)(2.5 \times 10^{-4}m)^2$$
$$A = \left(\frac{22}{7}\right)(6.25 \times 10^{-8}m)^2$$

$$R = \frac{(1.54 \times 10^{-8})(15)}{(1.963 \times 10^{-7})}$$
$$R = \frac{2.31 \times 10^{-7}}{(1.963 \times 10^{-7})}$$

$A = 1.963 \times 10^{-7}$

$\because R = \frac{\rho L}{A}$

$R = 1.176\,\Omega$

$\because V = IR$

$V = (4)(1.176)$

$V = 4.7\ Answer$

(iv) Find the Q-Value of the reaction

$${}_{94}Pu^{239} \rightarrow {}_{2}He^{4} + {}_{92}U^{22} + Q$$

The isotopic mass of Plutonium= 239.0522 u
The isotopic mass of Uranium= 235.0439 u
The isotopic mass of alpha particle= 4.0026 u

DATA:

$The\ isotopic\ mass\ of\ Plutonium = 239.0522\ u$
$The\ isotopic\ mass\ of\ Uranium = 235.0439\ u$
$The\ isotopic\ mass\ of\ alpha\ particle = 4.0026\ u$
$Mass\ of\ reactants = m_1 = 239.0522\ u$
$Mass\ of\ Product = m_2 = 235.0439 + 4.0026$
$m_2 = 239.0465\ u$
$\emptyset - value\ of\ reaction = ?$

SOLUTION:

$$Diff, of\ Mass = \Delta m = m_2 - m_1$$
$$\Delta m = 239.0465\ u - 239.0522u$$
$$\Delta m = 0.0057u$$
$$\emptyset - value\ Energy = \Delta m(931.5\ MeV)$$
$$\emptyset - value\ Energy = (0.0057)(931.5MeV)$$
$$\emptyset - value\ Energy = 5.3\ MeV\ Answer$$

(vi) A system absorbs 1147 joules of heat, loses 233 joules of heat by conduction to the surroundings and delivers 614 joules of work. Calculate the change in the internal energy of the system.

DATA:

$Q_1 = 1147\ J$
$Q_2 = 233\ J$
$\Delta W = 614\ J$
$\Delta U = ?$

SOLUTION:

$$\because \Delta Q = Q_2 - Q_1$$
$$\Delta Q = 1147 - 233 = 914\ J$$
$$\because \Delta U = \Delta Q - \Delta W$$
$$\Delta U = 914 - 614$$
$$\Delta U = 300\ J\ Answer$$

(viii) A photon of what minimum energy is required to excite a hydrogen atom from n=1 to n=3?($R_H = 1.097 \times 10^7 m^{-1}$)

DATA:

$n_i = 1$
$n_f = 3$
$R_H = 1.097 \times 10^7 m^{-1}$

$E = ?$

SOLUTION:

$$\because E = E_f' - E_i$$

$$E = \left(\frac{-13.6}{n_f^2}\right) - \left(\frac{-13.6}{n_i^2}\right)$$

$$E = \left(\frac{-13.6}{(3)^2}\right) - \left(\frac{-13.6}{(1)^2}\right)$$

$$E = \frac{-13.6}{9} + \frac{13.6}{1}$$

$$E = -1.5 + 13.6$$

$$E = -12.1\ eV \quad Answer$$

(x) Two unequal point charges repel each other by a force of 10 Newton's when they are 10 cm apart. Find the force which they exert on each other when they are 1 cm apart. If the magnitude of one point charge is $-4.25 \times 10^{-6}C$, find the magnitude of the other.

DATA:

$F_1 = 10\ N$

$r_1 = 10cm = \frac{10}{100} = 0.1m$

$r_2 = 1cm = \frac{1}{100} = 0.01m$

$q_1 = -4.25 \times 10^{-6}C$

$k = 9 \times 10^9$

$F_2 = ?$

$q_2 =$

SOLUTION:

$$\because F = k\,\frac{q_1 q_2}{r^2}$$

$$F_1 = k\,\frac{q_1 q_2}{{r_1}^2}$$

$$q_2 = \frac{F_1 {r_1}^2}{q_1\, k}$$

$$q_2 = \frac{(10)(0.1)^2}{(-4.25 \times 10^{-6})(9 \times 10^9)}$$

$$q_2 = \frac{0.1}{-38250}$$

$$q_2 = -2.614 \times 10^{-6}C$$

$$F_2 = k\,\frac{q_1 q_2}{{r_1}^2}$$

$$F_2 = \frac{(9 \times 10^9)(-4.25 \times 10^{-6})(-2.614 \times 10^{-6}C)}{(0.01)^2}$$

$$F_2 = \frac{0.0999}{0.0001}$$

$F_2 = 999 \approx 1000\,N$ *Answer*

(xiii) A current of 6.25 amperes is maintained in a long straight conductor by a source. Calculate the force pr meter on similar parallel conductor in air at a distance of 0.5 m from the first and carrying a current of 2 amperes. ($\mu_o = 4\pi \times 10^{-7} T.m / A$)

DATA:

$I_1 = 6.25\ Amperes$

$L = 1\,m$

$r = 0.5m$

$I_2 = 2\ Amperes$

$\mu_o = 4\pi \times 10^{-7}\ T.m/A$

$\theta = 90°$

$F = ?$

SOLUTION:

$$\because B = \frac{\mu_o I}{2\pi r}$$

$$B = \frac{\mu_o I_1}{2\pi r}$$

$$B = \frac{(4\pi \times 10^{-7})(6.25)}{2(\pi)(0.5)}$$

$$B = \frac{(4 \times 10^{-7})(6.25)}{1}$$

$$B = 2.5 \times 10^{-6}$$

$$F = I_2 LB\ Sin\,\theta$$

$$F = (2)(1)(2.5 \times 10^{-6})(sin\,90)$$

$$F = (5 \times 10^{-6})(1)$$

$$F = 5 \times 10^{-6}\ N \quad Answer$$

(xiv) A photon of wavelength 0.004 Å in the vicinity of a heavy nucleus produces an electron-positron pair. Find the kinetic energy of each particle of the pair in MeV, If the kinetic energy of positron is twice that of electron. ($m_oc^2 = 8.19 \times 10^{-14}\ joules, h = 6.63 \times 10^{-34} J.s\ and\ c = 3 \times 10^8\ m/s$)

DATA:

$\lambda = 0.004 Å = \frac{0.004}{10^{-10}} = 4 \times 10^{-13} m$

$m_oc^2 = 8.19 \times 10^{-14} joules$

$h = 6.63 \times\times 10^{-34} J.s$

$c = 3 \times 10^8\ m/s$

$K.E_{Positron} = (K.E)_{e+} = ?$

$K.E_{Electron} = (K.E)_{e-} = ?$

SOLUTION:

For pair production

$$hv = 2(m_oc^2)_e + (K.E)_{e-} + (K.E)_{e+}$$

$$Since\ (K.E)_{e+} = 2(K.E)_{e-}$$

$$hv = 2(m_oc^2)_e + (K.E)_{e-} + 2(K.E)_{e-}$$

$$v = \frac{c}{\lambda}$$

$$h\frac{c}{\lambda} = 2(m_oc^2)_e + 3(K.E)_{e-}$$

$$\frac{(6.63\times10^{-34})(3\times10^{8})}{4\times10^{-13}}=2(8.19\times10^{-14})+3(K.E)_{e-}$$
$$4.972\times10^{-34+8+13}=1.638\times10^{-13}+3(K.E)_{e-}$$
$$4.972\times10^{-13}=1.638\times10^{-13}+3(K.E)_{e-}$$
$$(4.972-1.638)\times10^{-13}=3(K.E)_{e-}$$
$$3(K.E)_{e-}=3.334\times10^{-13}$$
$$(K.E)_{e-}=\frac{3.334\times10^{-13}}{3}$$
$$(K.E)_{e-}=1.111\times10^{-13}\,J$$
$$\because 1\,eV=1.6021\times10^{-19}J$$
$$(K.E)_{e-}=\frac{1.111\times10^{-13}}{1.6021\times10^{-19}}$$
$$(K.E)_{e-}=6.937\times10^{5}\,ev$$
$$(K.E)_{e-}=\frac{0.6937\times10^{6}}{10^{6}}$$
$$(K.E)_{e-}=0.693MeV$$
$$But\ (K.E)_{e+}=2(K.E)_{e-}$$
$$(K.E)_{e+}=2(0.693MeV)$$
$$(K.E)_{e+}=1.386\,Mev$$

PHYSICS 2017

2(i)A particle of mass $1.67\times10^{-27}kg$ and charge 1.6×10^{-19}C is to be held motionless between two horizontal parallel plates, the voltage applied between the plates is 14.32×10^{-9}volt. Calculate the distance between the plates.

DATA:

$m=1.67\times10^{-27}kg$

$q=1.6\times10^{-19}C$

$\Delta V=14.32\times10^{-9}$

$d=?$

SOLUTION:

$$\because \Delta V=E\times d$$
$$d=\frac{\Delta V}{E}---(1)$$

In this case particle between two horizontal parallel plates is motionless, so

$$F_{Elec.}=W_{particle}$$
$$Eq=mg$$
$$E=\frac{mg}{q}$$
$$Ee=\frac{(1.67\times10^{-27})(9.8)}{1.6\times10^{-19}}$$
$$E=1.022\times10^{-26+19}$$
$$E=1.022\times10^{-7}\,Volt/meter$$

Substituting value of "E" in Equation (1)

$$d=\frac{14.32\times10^{-9}}{1.022\times10^{-7}}$$
$$d=13.99\times10^{-2}$$
$$d=0.1399\,meter\quad Answer$$

(iii) Calculate the temperature at which the root mean square speed of hydrogen molecules is 3300m/s. Give your answer in degree Celsius ($3.32 \times 10^{-27} kg$)

DATA:

$v_{r.m.s} = 3300\ m/s$
$m = 3.32 \times 10^{-27} kg$
$K = 1.38 \times 10^{-23} J/K$

SOLUTION:

$$\because v_{r.m.s} = \sqrt{\frac{3KT}{m}}$$

Squaring Both Sides

$$v_{r.m.s}^2 = \frac{3KT}{m}$$

$$(3300)^2 = \frac{3(1.38 \times 10^{-23})(T)}{(3.32 \times 10^{-27})}$$

$$1.089 \times 10^7 = \frac{4.14 \times 10^{-23} \cdot T}{(3.32 \times 10^{-27})}$$

$$T = \frac{(1.089 \times 10^7)(3.32 \times 10^{-27})}{4.14 \times 10^{-23}}$$

$$T = \frac{1.089 \times 3.32}{4.14} \times 10^{-27+7+23}$$

$$T = \frac{1.089 \times 3.32}{4.14} \times 10^{-27-7+23}$$

$$T = 0.8733 \times 10^3$$

$$T = 873.3\ K$$

$$T_c = 873.3 - 273$$

$$T_c = 600.3\ °C\ Answer$$

(v) A 400 volt meter has a total resistance of 40,000Ω. What additional series resistance must be connected to it to increase its range to 750volts?

DATA:

$V_1 = 400\ volts$
$R_v = 40,000\Omega$
$V_2 = 750\ volts$
$R_x =$

SOLUTION:

$$V_1 = IR_v$$

$$I = \frac{V_1}{R_x}$$

$$I = \frac{400}{40000} = 0.1\ Ampere$$

$$\because R_x = \frac{V_2}{I} - R_v$$

$$R_x = \frac{750}{0.1} - 40000$$

$$R_x = 75000 - 40000$$

$$R_x = 35000\Omega\ \ Answer$$

(vi) X-rays of wavelength 3.64×10^{-10} m are used in Compton scattering process. Find the fractional change in wavelength for a scattering angle of 120° (Give = $6.63 \times 10^{-34} Js, m_o = 9.1 \times 10^{-31} kg, c = 3 \times 10^8$)

DATA:

$\lambda_1 = 3.64 \times 10^{-10}$

$\theta = 120°$

$h = 6.63 \times 10^{-34} Js$

$m_o = 9.1 \times 10^{-31} kg$

$c = 3 \times 10^8 m/s$

fractional change $= \dfrac{\Delta\lambda}{\lambda_1} = ?$

SOLUTION:

$$\because \Delta\lambda = \frac{h}{m_o c}(1 - \cos\theta)$$

Divided by λ_1 *on both sides*

$$\frac{\Delta\lambda}{\lambda_1} = \frac{h}{m_o c \lambda_1}(1 - \cos\theta)$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{6.63 \times 10^{-34}}{(9.1 \times 10^{-31})(3 \times 10^{8})(3.64 \times 10^{-10})}(1 - \cos 120)$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{6.63 \times 10^{-34}}{9.1 \times 3 \times 3.64 \times 10^{-31+8-10}}(1 + 0.5)$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{9.945 \times 10^{-34}}{9.9372 \times 10^{-32}}$$

$$\frac{\Delta\lambda}{\lambda_1} = 1.0 \times 10^{-2}$$

$$\frac{\Delta\lambda}{\lambda_1} = \frac{0.010}{100}$$

$$\frac{\Delta\lambda}{\lambda_1} = 1\% \; Answer$$

(vii)The half-lifeof $_{104}Po^{210}$ is 140 days. By what percent will its activity decrease per hour?

DATA:

$T_{\frac{1}{2}} = 140\ day = 140 \times 24 = 3360\ hours$

$\lambda = ?$

SOLUTION:

$$\because T_{\frac{1}{2}}\lambda = 0.693$$

$$\lambda = \frac{0.693}{T_{\frac{1}{2}}}$$

$$\lambda = \frac{0.693}{3360}$$

$$\lambda = 2.063 \times 10^{-4}$$

For Percentage

$$\lambda = 2.063 \times 10^{-4} \times 100$$

$$\lambda\ in\ percent = 0.021\ \%$$

(viii)How much energy is needed to ionize a hydrogen atom originally in ground state?

(Give $h = 6.63 \times 10^{-34} Js, c = 3 \times 10^{8}\ m/s R_H = 1.097 \times 10^{7}/m$)

DATA:

$n_i = 1$

$n_f = \infty$

$h = 6.63 \times 10^{-34} Js$

$c = 3 \times 10^{8}\ m/s$

$R_H = 1.097 \times 10^{7}/m$

$E = ?$

SOLUTION:

$$\because \frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda} = (1.097 \times 10^{7})\left(\frac{1}{(1)^2} - \frac{1}{(\infty)^2}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^{7}\left(\frac{1}{1} - \frac{1}{\infty}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^{7}(1-0)$$

$$\frac{1}{\lambda} = 1.097 \times 10^{7}$$

$$\lambda = \frac{1}{1.097 \times 10^{7}}$$

$$\lambda = 9.116 \times 10^{-8} Meter$$

Energy needed for Ionization of Hydrogen atom will be

$$\because E = hv$$

$$But\ \ v = \left(\frac{c}{\lambda}\right)$$

$$E = \frac{hc}{\lambda}$$

$$E = \frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{(9.116 \times 10^{-8})}$$

$$E = \frac{19.89}{9.116} \times 10^{-34+8+8}$$

$$E = \frac{19.89}{9.116} \times 10^{-34+8+8}$$

$$E = 2.181 \times 10^{-18} Joules$$

OR

$$E = \frac{2.181 \times 10^{-18}}{1.6 \times 10^{-19}}$$

$$E = 13.6\ e.v Answer$$

(x) An alternating current generator operate at 79Hz. The area of the coil is $500 cm^2$. Calculate the number of turns in the coil when a magnetic field induction 0.06 tesla produces a maximum potential difference of 149volts.

DATA:

$f = 79\ Hz$

$A = 500\ cm^2 = \frac{500}{100 \times 100} = 5 \times 10^{-2} m^2$

$B = 0.06\ tesla$

$V = 149\ volts$

$N = ?$

SOLUTION:

$$\because V = BNA\omega$$
$$But\ \omega = 2\pi f$$
$$\therefore V = BNA(2\pi f)$$
$$N = \frac{V}{2BA\pi f}$$
$$N = \frac{149}{2(0.06)(5\times10^{-2})\left(\frac{22}{7}\right)(79)}$$
$$N = \frac{149}{148.971}\times10^2$$
$$N = \frac{149}{148.971}\times10^2$$
$$N = 100\ Turns\ \ Answer$$

(xi) A deuteron of mass 3.3431×10^{-27}kg is formed when a proton of mass 1.6724×10^{-27} kg and neutron of mass 1.6748×10^{-27}kg combine. Calculate the mass defect and binding energy in MeV.

DATA:

$m_d = 3.3431\times10^{-27}$kg

$m_p = 1.6724\times10^{-27}$

$m_n = 1.6748\times10^{-27}$

$\Delta m = ?$

$B.E = ?$

SOLUTION:

$$\Delta m = m_p + m_n - m_d$$
$$\Delta m = 1.6724\times10^{-27} + 1.6748\times10^{-27} - 3.3431\times10^{-27}$$
$$\Delta m = (1.6724 + 1.6748 - 3.3431)\times10^{-27}$$
$$\Delta m = (3.3472 - 3.3431)\times10^{-27}$$
$$\Delta m = 0.0041\times10^{-27}$$
$$\Delta m = 4.1\times10^{-30}kg$$
$$B.E = \Delta mc^2$$
$$B.E = (4.1\times10^{-30})(3\times10^8)^2$$
$$B.E = (4.1\times10^{-30})(9\times10^{16})$$
$$B.E = 3.69\times10^{-13}J$$
$$B.E = \frac{3.69\times10^{-13}}{1.6\times10^{-19}}$$
$$B.E = 2.30625\times10^7\ ev$$
$$B.E = \frac{2.30625\times10^7}{10^6}$$
$$B.E = 2.30625\ Mev\ \ Answer$$

(xii) The resistance of a platinum resistance thermometer is 200 ohms at 0°C and 257.6 ohms when immersed in a hot bath . what is the temperature of the bath(α=0.00392/°C)

$R_o = 200\ ohm$

$R_t = 257.6\ ohm$

$t = ?$

$\alpha = 0.00392/oC$

SOLUTION:

$$\because R_t = R_o(1 + \alpha t)$$
$$257.6 = 200(1 + 0.00392t)$$
$$\frac{257.6}{200} = 1 + 0.00392t$$
$$0.00392t = 1.288 - 1$$
$$t = \frac{0.288}{0.00392}$$
$$t = 73.47\,°C \quad Answer$$

PHYSICS 2016

2(i) The high temperature reservoir of a Carnot engine is at 200°C and has an efficiency of 35% to increase the efficiency to 45% by how many degrees should the temperature of cold reservoir be decreased if the temperature of the high temperature reservoir remains constant?

DATA:

$T_1 = 200°C = 200 + 273 = 473K$

$E_1 = 35\,\%$

$E_2 = 45\,\%$

$\Delta T_2 = ?$

SOLUTION:

$$\because E_1 = 1 - \frac{T_2}{T_1}$$
$$\frac{T_2}{T_1} = 1 - E_1$$
$$\frac{T_2}{473} = 1 - \frac{35}{100}$$
$$T_2 = (1 - 0.35) \times 473$$
$$T_2 = 0.65 \times 473$$
$$T_2 = 307.45\,K$$

Temperature of cold reservoir at

final Efficiency

$$E_2 = 1 - \frac{T_2'}{T_1}$$
$$\frac{T_2'}{T_1} = 1 - E_2$$
$$\frac{T_2'}{473} = 1 - \frac{45}{100}$$
$$T_2 = (1 - 0.45) \times 473$$
$$T_2 = 0.55 \times 473$$
$$T_2 = 2.60.15\,K \quad Answer$$

(iv) find the resistance at 100°C of a Silver wire, 1 mm in diameter and 1000 cm long.

DATA:

$T = 100°C$

$D = 1mm = \frac{1}{1000} = 0.001m$

$r = \frac{D}{2} = \frac{0.001}{2} = 0.0005$

$L = 1000cm = \frac{1000}{100} = 10m$

$\alpha_{sliver} = 0.0038\ /C.$

$\because \rho_{sliver} = 1.52 \times 10^{-8}\Omega m$

$R_t = R_{100} = ?$

SOLUTION:

$$R_t = R_o(1 + \alpha t) - - - (1)$$

$$R_o = \rho\frac{L}{A}$$
$$\because A = \pi r^2$$
$$R_o = \frac{\rho L}{\pi r^2}$$
$$R_o = \frac{(1.52 \times 10^{-8})(10)}{\left(\frac{22}{7}\right)(0.0005)^2}$$
$$R_o = \frac{1.52 \times 10^{-7}}{(3.142)(2.25 \times 10^{-7})}$$
$$R_o = \frac{1.52}{7.885} \times 10^{-7+7}$$
$$R_o = 0.193\Omega$$

Now

$$R_{100} = (0.193)[1 + (0.0038)(100)]$$
$$R_{100} = (0.193)(1 + 0.38)$$
$$R_{100} = (0.193)(1.38)$$
$$R_{100} = 0.266\Omega \quad Answer$$

(v) An iron core solenoid with 600 turns has a cross section area of 2.0 cm². A current of 4.0 ampere passing through it produces B=0.4 weber/m². What emf is produced in it, if the current is turned off in 0.2 second? What is its self-inductance?

DATA:

$N = 600$

$A = 2.0\ \text{cm}^2 = \frac{2}{100 \times 100} = 2 \times 10^{-4} m^2$

$\Delta I = (4.0 - 0) = 4.0\ Ampere$

$B = 0.4 \text{weber/m}^2$

$e.m.f. = \xi =$

$\Delta t = 0.2\ second$

Self Inductance $= L = ?$

SOLUTION:

$$\because \xi = N.\frac{\Delta\phi}{\Delta t}$$
$$But\ \Delta\phi = B.\Delta A$$
$$\therefore \xi = N.\frac{B.\Delta A}{\Delta t}$$
$$\xi = (600)\frac{(0.4)(2 \times 10^{-4})}{(0.2)}$$
$$\xi = \frac{480}{0.2} \times 10^{-4}$$
$$\xi = 0.24\ Volts$$

$$\because \xi = L\frac{\Delta I}{\Delta t}$$
$$L = \frac{\xi\, \Delta t}{\Delta I}$$
$$L = \frac{(0.24)(0.2)}{(4.0)}$$
$$L = \frac{0.048}{4.0}$$
$$L = 0.012\ Henry$$
$$L = 0.012 \times 1000$$
$$L = 12\ mH \quad Answer$$

(vi) Sodium surface is shown with light of wavelength $3 \times 10^{-7} m$. Find the kinetic energy of the emitted photo electron and the cut-off wavelength of Sodium. Work function of sodium is 2.46 eV.

DATA:

$\lambda = 3 \times 10^{-7} m$

$\phi_o = 2.46\, eV = 2.46 \times 1.6 \times 10^{-19}$

$\phi_o = 3.936 \times 10^{-19} Joule$

$h = 6.63 \times 10^{-34} J.S$

$K.E_{max} = ?$

$\lambda_c = ?$

SOLUTION:

$$\because \frac{hc}{\lambda} = \phi_o + K.E_{max}$$

$$\frac{hc}{\lambda} = \phi_o + K.E_{max}$$

$$K.E_{max} = \frac{hc}{\lambda} - \phi_o$$

$$K.E_{max} = \frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{3 \times 10^{-7}} - 3.936 \times 10^{-19}$$

$$K.E_{max} = 6.63 \times 10^{-19} - 3.936 \times 10^{-19}$$

$$K.E_{max} = (6.63 - 3.936) \times 10^{-19}$$

$$K.E_{max} = 2.694 \times 10^{-19} J$$

$$K.E_{max} = \frac{2.694 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$K.E_{max} = 1.684\, eV$$

$$\because \lambda_c = \frac{hc}{\phi}$$

$$\lambda_c = \frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{3.936 \times 10^{-19}}$$

$$\lambda_c = \frac{6.63 \times 3}{3.936} \times 10^{-34+8+19}$$

$$\lambda_c = 5.0533 \times 10^{-7} m$$

$$\lambda_c = 5.0533 \times 10^{-7} \times 10^{10}$$

$$\lambda_c = 5.0533 \times 10^{3}$$

$$\lambda_c = 5053.3\, \dot{A}$$

(vii) Find the shortest and longest wavelength of emitted photons in Hydrogen spectra in Pfund series.

DATA:

$\lambda_{Shortest} = ?$

$\lambda_{Longest} = ?$

For Shortest wavelength $n_i = \infty$

for longest wavelength $n_i' = 6$

$n_f = 5$

SOLUTION:

FOR SHORTEST WAVELENGTH

$$\because \frac{1}{\lambda_{Shortest}} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda_{Shortest}} = (1.097 \times 10^7)\left(\frac{1}{(5)^2} - \frac{1}{(\infty)^2}\right)$$

$$\frac{1}{\lambda_{Shortest}} = 1.097 \times 10^7\left(\frac{1}{25} - 0\right)$$

$$\frac{1}{\lambda_{Shortest}} = 1.097 \times 10^7\left(\frac{1}{25}\right)$$

$$\frac{1}{\lambda_{Shortest}} = 4.388 \times 10^5$$

$$\lambda_{Shortest} = \frac{1}{4.388} \times 10^{-5}$$

$$\lambda_{Shortest} = 2.279 \times 10^{-6} Meter$$

FOR LONGEST WAVELENGTH

$$\because \frac{1}{\lambda_{longest}} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda_{longest}} = (1.097 \times 10^7)\left(\frac{1}{(5)^2} - \frac{1}{(6)^2}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.097 \times 10^7\left(\frac{1}{25} - \frac{1}{36}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.097 \times 10^7\left(\frac{36 - 25}{900}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.097 \times 10^7\left(\frac{11}{900}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.34 \times 10^5$$

$$\lambda_{longest} = \frac{1}{1.34} \times 10^{-5}$$

$$\lambda_{longest} = 7.465 \times 10^{-6} Meter \quad Answer$$

(viii) The number of atoms per gram of $^{226}_{88}Ra$ is 2.666×10^{21} and it decays with a half life of 1622 years. Find the activity and decay constant of the sample.

DATA:

$^{226}_{88}Ra = N = 2.666 \times 10^{21} atoms$

$T_{1/2} = 1622 \times 365 \times 24 \times 60 \times 60 = 5.115 \times 10^{10} seconds$

$\lambda = ?$

$R = ?$

SOLUTION:

$$T_{1/2} = \frac{0.693}{\lambda}$$
$$\lambda = \frac{0.693}{T_{1/2}}$$
$$\lambda = \frac{0.693}{5.115 \times 10^{10}}$$
$$\lambda = 1.345 \times 10^{-11} \text{ per Second}$$

$$\frac{\Delta N}{\Delta t} = \lambda N$$
$$\text{But } R = \frac{\Delta N}{\Delta t}$$
$$R = (1.345 \times 10^{-11})(2.666 \times 10^{21})$$

$$R = 3.5859 \times 10^{10} \text{Disintegration / sec. } Answer$$

(ix) An α-particle of charge 3.2×10^{-19}C and mass 6.68×10^{-27}kg is held motionless between two horizontal parallel plats separated by 10 cm. Find the potential difference between the plates.

DATA:

$m = 6.68 \times 10^{-27} kg$

$q = 3.2 \times 10^{-19} C$

$d = 10\ cm = 0.1 m$

$\Delta V = ?$

SOLUTION:

$$\because \Delta V = E \times d - - - (1)$$

In this case particle between two horizontal parallel plates is motionless, so

$$F_{Elec} = W_{particle}$$
$$Eq = mg$$
$$E = \frac{mg}{q}$$
$$Ee = \frac{(6.68 \times 10^{-27})(9.8)}{3.2 \times 10^{-19}}$$
$$E = 20.547 \times 10^{-27+19}$$
$$E = 2.0547 \times 10^{-7} \, Volt/meter$$

Substituting value of "E" in Equation (1)

$$\Delta V = 2.0547 \times 10^{-7} \times 0.1$$
$$\Delta V = 2.055 \times 10^{-8} volt \quad Answer$$

(xi) A proton of charge 1.6×10^{-19}C and mass 1.67×10^{-27} kg is accelerated by a potential difference of 6×10^5 volts. Then its enters perpendicularly into a magnetic field of intensity 0.5 Tesla. Find the radius of the circular path of the proton.

DATA:

$e = 1.6 \times 10^{-19} C$

$m = 1.67 \times 10^{-27}\ kg$

$V = 6 \times 10^5\ volts$

$\theta = 90°$

$B = 0.5\ Tesla$

$r = ?$

SOLUTION:

$$\because \frac{1}{2}mv^2 = V.e$$
$$v^2 = \frac{2V.e}{m}$$
$$v = \sqrt{\frac{2V.e}{m}}$$
$$v = \sqrt{\frac{2\,(6 \times 10^5)(1.6 \times 10^{-19})}{1.67 \times 10^{-27}}}$$
$$v = \sqrt{\frac{19.2}{1.67} \times 10^{5-19+27}}$$
$$v = \sqrt{1.149 \times 10^{14}}$$
$$v = 1.072 \times 10^7$$

$$\because B.e.v\sin\theta = \frac{mv^2}{r}$$
$$r = \frac{mv^2}{B.e.v\sin\theta}$$
$$r = \frac{mv}{B.e.\sin\theta}$$
$$r = \frac{(1.67 \times 10^{-27})(1.072 \times 10^7)}{(0.5)\,(1.6 \times 10^{-19})\,(\sin 90)}$$
$$r = \frac{1.790}{0.8} \times 10^{-27+7+19}$$
$$r = 0.22375\ cm$$
OR
$$r = 22.375\ m \quad Answer$$

(xiv) Find the binding energy and the packing fraction in MeV of ${}_{52}Te^{126}$ given that $m_p = 1.0078\ amu$, $m_n = 1.0086\ amu$, $m_{Te} = 125.9033\ amu$ and 1u=931.5MeV

DATA:
$m_p = 1.0078\ amu$
$m_n = 1.0086\ amu$
$m_{Te} = 125.9033\ amu$
$A = 126$
$E = ?$
$f = ?$

SOLUTION:
$No.\ of\ proton = 52$
$No.\ of\ Neutrons = 126 - 52 = 74$
$Mass\ of\ 52\ Proton = 52 \times 1.0078 = 52.4056$
$Mass.\ of\ Neutrons = 74 \times 1.0086 = 74.6364$
$Total\ Mass.\ of\ Proton\ and\ Neutrons = 52.4056 + 74.6364$
$Total\ Mass.\ of\ Proton\ and\ Neutrons = 127.042\ amu$

$$\Delta m = Total\ Mass.\ of\ Proton\ and\ Neutrons - mass\ of\ {}_{52}Te^{126}$$
$$\Delta m = 127.042 - 125.9033$$
$$\Delta m = 1.1387\ amu$$
$$But\ 1amu = 1.66 \times 10^{-27} kg$$
$$\Delta m = 1.1387 \times 1.66 \times 10^{-27}$$
$$\Delta m = 1.890 \times 10^{-27} kg$$

FOR BINDING ENERGY:

$$\because Binding\ Energy = E = \Delta mc^2$$
$$E = (1.890 \times 10^{-27})(3 \times 10^8)^2$$
$$E = (1.890 \times 10^{-27})(9 \times 10^{16})$$
$$E = 1.701 \times 10^{-10} J$$

$$E = \frac{1.701 \times 10^{-10}}{1.6021 \times 10^{-19}}$$
$$E = \frac{1.701}{1.6021} \times 10^{-10+19}$$
$$E = 1.061731 \times 10^{9}\ ev$$
$$E = \frac{1.061731 \times 10^{9}\ ev}{10^{6}}$$
$$E = 1061.731\ Mev$$

PACKING FRACTION

$$\because f = \frac{\Delta m}{A}$$
$$f = \frac{1.1387}{126}$$
$$f = 0.00937\ \ Answer$$

PHYSICS 2015

(iv) What will be the relativistic velocity of a particle whose kinetic energy is twice of its rest mass energy?

DATA:

$K.E = 2m_oc^2$

$T.E. = E_o + K.E.$

SOLUTION:

$$mc^2 = m_oc^2 + 2m_oc^2$$
$$mc^2 = 3m_oc^2$$
$$m = 3m_o$$
$$\because m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$$
$$3m_o = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$$
$$3 = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

Squaring both sides

$$9 = \frac{1}{1-\frac{v^2}{c^2}}$$
$$1-\frac{v^2}{c^2} = \frac{1}{9}$$
$$1-\frac{1}{9} = \frac{v^2}{c^2}$$
$$\frac{9-1}{9} = \frac{v^2}{c^2}$$
$$\frac{8}{9} = \frac{v^2}{c^2}$$
$$v^2 = \frac{8}{9}c$$
$$v = \frac{2}{3}\sqrt{2}\,c\ \ Answer$$

Relativistic velocity of a particle whose kinetic energy is twice of its rest mass energy is $\frac{2}{3}\sqrt{2}\,c$

(v) Find the Binding energy and Packing fraction(B.E. per nucleon) of $_{52}Te^{126}$. Given that $m_p = 1.0078U, m_n = 1.0086U, m_{Te} = 125.9033U IU = 931.5MeV$

(See Question No. 2 (xiv), 2016)

(vi) A heat engine performs 200 J of work in each cycle and has an efficiency of 30 percent. For each cycle of operation, (a)how much heat is absorbed? (b) how much heat is expelled?

DATA:

$\Delta W = 200\,J$

$E = 30\%$

$Q_1 = ?$

$Q_2 = ?$

SOLUTION:

$$\because E = 1 - \frac{Q_2}{Q_1}$$

$$E = \frac{Q_1 - Q_2}{Q_1}$$

$$But\ \Delta W = Q_1 - Q_2$$

$$E = \frac{\Delta W}{Q_1}$$

$$\frac{30}{100} = \frac{200}{Q_1}$$

$$Q_1 = 200 \times \frac{10}{3}$$

$$Q_1 = 666.66\,J$$

$$\Delta W = Q_1 - Q_2$$

$$Q_2 = Q_1 - \Delta W$$

$$Q_2 = 666.66 - 200$$

$$Q_2 = 466.66\,J\ Answer$$

(vii) A 50 ohm resistor is to be wound from a platinum wire 0.1 mm in diameter. How much wire is needed?

DATA:

$R = 50\,ohm$

$D = 0.1mm = \frac{0.1}{100} = 0.1 \times 10^{-3} m$

$R = \frac{D}{2} = \frac{0.1 \times 10^{-3}}{2} m = 5 \times 10^{-5} m$

$\rho = 11 \times 10^{-8}\,ohm.m$

$L = ?$

SOLUTION:

$$\because A = \pi r^2$$

$$A = \left(\frac{22}{7}\right)(5 \times 10^{-5})^2$$

$$A = (3.142)(2.5 \times 10^{-7})$$

$$A = 7.853 \times 10^{-9}\,m^2$$

$$\because R = \rho\frac{L}{A}$$

$$L = \frac{RA}{\rho}$$

$$L = \frac{(50)(7.853 \times 10^{-9})}{(11 \times 10^{-8})}$$

$$L = \frac{392.65}{11} \times 10^{-9+8}$$

$$L = 3.569\,m$$

(viii) A galvanometer, whose resistance is 60 ohms, deflects full scale for a potential diff. of 100 millivolts across its terminals. What shunt resistance must be connected to convert it into an ammeter of 5 ampere range?

DATA:

$R_g = 60\ ohm$

$V_g = 100mv = \frac{100}{1000} = 0.1\ volts$

$I = 5\ ampere$

$R_s = ?$

SOLUTION:

$$\because V_g = I_g R_g$$

$$I_g = \frac{V_g}{R_g}$$

$$I_g = \frac{0.1}{60}$$

$$= 1.666 \times 10^{-3}\ Ampere$$

$$\because R_s = \frac{I_g R_g}{I - I_g}$$

$$R_s = \frac{(1.666 \times 10^{-3})(60)}{5 - 1.67 \times 10^{-3}}$$

$$R_s = \frac{0.0999}{5 - 0.001666}$$

$$R_s = \frac{0.0999}{5 - 0.00167}$$

$$R_s = \frac{0.0999}{4.998}$$

$$R_s = 0.01998\ Ohms \quad Answer$$

(ix) An e.m.f. of 45 millivolts is induced in a coil of 500 turns. When the current in a neighboring coil changes form 15 amps to 4 amps in 0.2 seconds,

(a) What is the mutual inductance of the coils?

(b) What is the rate of change of flux in the second coil?

DATA:

e.m.f. $= \xi_s = 45$ millivolts $= \frac{45}{1000} = 0.045$ Volts

$N = 500\ turns$

$I_1 = 15\ Amperes$

$I_2 = 4\ Amperes$

$\Delta I_p = I_1 - I_2 = 15 - 4 = 11\ Amperes$

$\Delta t = 0.2\ seconds$

mutual inductance $= M = ?$

$Rate\ of\ change\ of\ Flux\ in\ second\ coil = \frac{\Delta \phi}{\Delta t} = ?$

SOLUTION:

$$\because \xi_s = \frac{M\,\Delta I_p}{\Delta t}$$

$$M = \frac{\xi_s \Delta t}{\Delta I_p}$$

$$M = \frac{(0.045)(0.2)}{(11)}$$

$$M = 8.1818 \times 10^{-4}\ Henry$$

$$\because \xi_s = N_s \cdot \frac{\Delta \phi_s}{\Delta t}$$

$$\frac{\Delta\phi_s}{\Delta t} = \frac{\xi_s}{N_s}$$

$$\frac{\Delta\phi_s}{\Delta t} = \frac{0.045}{500}$$

$$\frac{\Delta\phi_s}{\Delta t} = 9\times10^{-5}\ weber/second \quad Answer$$

(x) A thin infinite sheet of uniformly distributed positive charge attracts a light sphere having a charge $-5\times10^{-6}C$ with a force of 1.695 N. Calculate the surface charge density of the sheet ($\varepsilon_o = 8.85\times10^{-12}C^2/Nm^2$).

DATA:

$|q| = 5\times10^{-6}C$

$F = 1.695N$

$\varepsilon_o = 8.85\times10^{-12}C^2/Nm^2$

$\sigma = ?$

SOLUTION:

$$\because E = \frac{\sigma}{2\varepsilon_o}$$

$$\frac{F}{q} = \frac{\sigma}{2\varepsilon_o}$$

$$\sigma = \frac{2\varepsilon_o F}{q}$$

$$\sigma = \frac{2(8.85\times10^{-12})(1.695)}{(5\times10^{-6})}$$

$$\sigma = 610^{-6}c/m^2 \quad Answer$$

(xi) Find the shortest wavelength of photon emitted in the Balmer series and determine its energy in eV. ($R_H = 1.097\times10^7m^{-1}$)

DATA:

$n_i = \infty$

$n_f = 2$

$h = 6.63\times10^{-34}Js$

$c = 3\times10^8\ m/s$

$R_H = 1.097\times10^7/\text{m}$

$E = ?$

SOLUTION:

$$\because \frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda} = (1.097\times10^7)\left(\frac{1}{(2)^2} - \frac{1}{(\infty)^2}\right)$$

$$\frac{1}{\lambda} = 1.097\times10^7\left(\frac{1}{4} - \frac{1}{\infty}\right)$$

$$\frac{1}{\lambda} = 1.097\times10^7\left(\frac{1}{4} - 0\right)$$

$$\frac{1}{\lambda} = 1.097\times10^7\left(\frac{1}{4}\right)$$

$$\frac{1}{\lambda} = 2.742\times10^6$$

$$\lambda = \frac{1}{2.742 \times 10^{6}}$$

$$\lambda = 3.647 \times 10^{-7} Meter$$

Energy needed for Ionization of Hydrogen atom will be

$$E = hv$$

$$But\ v = \left(\frac{c}{\lambda}\right)$$

$$E = \frac{hc}{\lambda}$$

$$E = \frac{(6.63 \times 10^{-34})(3 \times 10^{8})}{(3.647 \times 10^{-7})}$$

$$E = \frac{19.89}{3.647} \times 10^{-34+8+7}$$

$$E = 5.453 \times 10^{-19} Joules$$

$$E = \frac{5.453 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$E = 3.408\ e.v Answer$$

(xii) Calculate the speed of the electromagnetic wave, given that, $\varepsilon_o = 8.85 \times 10^{-12} C^2/Nm^2, \mu_o = 4\pi \times 10^{-7} web/Am$

DATA:

$\varepsilon_o = 8.85 \times 10^{-12} C^2 / Nm^2$

$\mu_o = 4\pi \times 10^{-7} web/Am$

$C = ?$

SOLUTION:

$$\because C = \frac{1}{\sqrt{\varepsilon_0 \mu_o}}$$

$$C = \frac{1}{\sqrt{(8.85 \times 10^{-12})(4\pi \times 10^{-7})}}$$

$$C = \frac{1}{\sqrt{(8.85 \times 4\pi) \times 10^{-12-7}}}$$

$$C = \frac{1}{\sqrt{(111.212) \times 10^{-19}}}$$

$$C = \frac{1}{\sqrt{1.1121 \times 10^{-17}}}$$

$$C = \frac{1}{3.33 \times 10^{-9}}$$

$$C = 29.98 \times 10^{8}\ Answer$$

(xiv) How many electrons should be removed from each of the two similar spheres, each of mass 10g so that electrostatic repulsion is balanced by the gravitational force? (Gravitationnel constant $G = 6.67 \times 10^{-11}\ Nm^2 / kg^2 and K = 9 \times 10^9 Nm^2/C^2$)

DATA:

$m_1 = m_2 = m = 10\ g = 0.01\ Kg$

$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$

$K = 9 \times 10^9 Nm^2/C^2$

$q_1 = q_1 = q$

$n = ?$

SOLUTION:

According to the given condition

$$\because F_{Gravitation} = F_{electrostatic}$$

$$q^2 = (7.411 \times 10^{-5}) \times 10^{-20}$$

$$G\frac{m_1m_2}{r^2} = K\frac{q_1q_2}{r^2}$$
$$Gm^2 = Kq^2$$
$$q^2 = G\frac{m^2}{K}$$
$$q^2 = (6.67\times10^{-11})\frac{(0.01)^2}{(9\times10^9)}$$
$$q^2 = \left(\frac{6.67\times0.0001}{9}\right)\times10^{-11-9}$$
$$q^2 = 7.411\times10^{-25}$$
$$q = 8.608\times10^{-13}\ Coulomb$$
$$Total\ charge = q = ne$$
$$n = \frac{q}{e}$$
$$n = \frac{8.608\times10^{-13}}{1.6\times10^{-19}}$$
$$n = 5.380\times10^6\ Electron$$

PHYSICS 2014

2(i) A galvanometer, having resistance 50Ω, deflects full scale for a potential difference of 100 mV across the terminals. What resistance should be connected to increase its range to 50 Volts?

DATA:

$R_g = 50\ ohm$

$V_g = 100mv = \frac{100}{1000} = 0.1\ volts$

$V = 50\ Volts$

$R_x = ?$

SOLUTION:

$$\because V_g = I_gR_g$$
$$I_g = \frac{V_g}{R_g}$$
$$I_g = \frac{0.1}{50}$$
$$I_g = 0.002\ Ampere$$
$$\because R_x = \frac{V}{I_g} - R_g$$
$$R_x = \frac{50}{0.002} - 50$$
$$R_x = 25000 - 50$$
$$R_x = 24950\ ohms \quad Answer$$

(ii) A rectangular bar of iron is 2cm × 2cm in cross section and 20 cm long. What will be its resistance at 500°C ?if $\alpha = 0.0052k^{-1}$ and $\rho = 11\times10^{-8}\Omega m$.

DATA:

$A = 2cm\times2cm = 4cm = \frac{4}{10000} = 0.0004m^2$

$L = 20cm = \frac{20}{100} = 0.2m$

$t = 500°C$

$\alpha = 0.0052k^{-1}$

$\rho = 11\times10^{-8}\Omega m$

$\because R_t = R_{500} = ?$

SOLUTION:

$$\because R_t = R_o(1+\alpha t) --- (1)$$

$$R_o = \rho\frac{L}{A}$$

$$R_o = \frac{(11\times10^{-8})(0.2)}{0.0004}$$

$$R_o = \frac{2.22\times10^{-8}}{4\times10^{-4}} = \frac{2.22}{4}\times10^{-8+4}$$

$$R_o = \frac{2.22\times10^{-8}}{4\times10^{-4}} = \frac{2.22}{4}\times10^{-8+4}$$

$$R_o = 5.5\times10^{-5}\Omega$$

Now

$$R_{500} = (5.5\times10^{-5})[1+(0.0052)(500)]$$

$$R_{500} = (5.55\times10^{-5})(1+2.6)$$

$$R_{500} = (5.55\times10^{-5})(3.6)$$

$$R_{500} = 1.998\times10^{-4}\Omega \ \ Answer$$

(iii) An iron core solenoid with 500 turns has a cross section of 5 cm^2.A current of 2.3 ampere passing through it produces of flux of B=0.53 Tesla. How large an e.m.f. is induced in it, if the current is turned off in 0.1 second?

DATA:

$N = 500\ Turns$

$A = 5cm^2 = \frac{5}{100\times100} = 0.0005\ m^2$

$\Delta I = 2.3 - 0 = 2.3\ Ampare$

$\Delta B = 0.53\ Tesla$

$\Delta t = 0.1\ second$

$e.m.f.Induced = E = ?$

$Self\ Inductance\ of\ Solenoid = L = ?$

SOLUTION:

$$\because \Delta B = \frac{\Delta\phi}{A}$$

$$\Delta\phi = \Delta B\ A$$

$$\Delta\phi = (0.53)(0.0005)$$

$$\Delta\phi = 2.65\times10^{-4}\ weber$$

$$\because E = N\frac{\Delta\phi}{\Delta t}$$

$$E = (500)\frac{(2.65\times10^{-4})}{(0.1)}$$

$$E = 1.325\ volts$$

$$\because E = L\frac{\Delta I}{\Delta t}$$

$$L = \frac{E\,\Delta t}{\Delta I}$$

$$L = \frac{(1.325)(0.1)}{(2.3)}$$

$$L = 0.0576\ Hertz$$

$$L = \frac{0.0576}{1000}\ Hertz$$

$$L = 576\ m\ Hertz \quad Answer$$

(iv) What will be the velocity and momentum of particle whose mass rest is m_o and kinetic energy is equal to twice of its rest mass energy.

(See Question No. 2 (iv) 2015)

(v) Find the Binding energy and packing fraction in MeV of $_{52}Te^{126}$ given that $m_p = 1.0078u$, $m_n = 1.0086u$, $m_{Te} = 125.9033u$ and $1U = 931.5MeV$

(See Question No. 2 (xiv), 2016)

(vi) Three resistors each of 50 Ω can be connected in four different ways. Find the equivalent resistance for each combination.

DATA:

$R_1 = R_2 = R_3 = 50\,\Omega$

$R_e = ?$

SOLUTION:

1st way:

Connected in series

$$\because R_e = R_1 + R_2 + R_3$$
$$R_e = 50 + 50 + 50$$
$$R_{e1} = 150\,\Omega$$

2nd way:

Connected in parallel

$$\because \frac{1}{R_{e2}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$
$$\frac{1}{R_{e2}} = \frac{1}{50} + \frac{1}{50} + \frac{1}{50}$$
$$\frac{1}{R_{e2}} = \frac{3}{50}$$
$$R_{e2} = \frac{50}{3} = 16.66\Omega$$

3rd way:

Connected two in series and one in parallel

$$R_e' = R_1 + R_2$$
$$R_e' = 50 + 50$$
$$R_e' = 100\Omega$$
$$\frac{1}{R_{e3}} = \frac{1}{R_e'} + \frac{1}{R_3}$$
$$\frac{1}{R_{e3}} = \frac{1}{100} + \frac{1}{50}$$
$$\frac{1}{R_{e3}} = \frac{1+2}{100}$$
$$R_{e3} = \frac{100}{3}$$
$$R_{e3} = 33.33\Omega$$

4th way:

Connected two in parallel and one in series

$$\because \frac{1}{R_e''} = \frac{1}{R_1} + \frac{1}{R_2}$$
$$\frac{1}{R_e''} = \frac{1}{50} + \frac{1}{50}$$
$$\frac{1}{R_e''} = \frac{1}{50} + \frac{1}{50}$$
$$\frac{1}{R_e''} = \frac{2}{50}$$
$$R_e'' = \frac{50}{2}$$
$$R_e'' = 25\Omega$$
$$\because R_{e4} = R_e'' + R_3$$
$$R_{e4} = 25 + 50$$
$$R_{e4} = 75\,\Omega \quad Answer$$

(vii) Calculate root mean square speed of Oxygen molecule at 800K. Its molar mass is 32 gm and universal gas constant R=8.31j/mole-K

DATA:

$T = 800K$

$M = 32g = \frac{32}{1000} = 0.032Kg$

$\mathrm{R} = 8.31\mathrm{j/mole - K}$

$\mathrm{K} = 1.38 \times 10^{-23}\, J/K$

SOLUTION:

$$\text{Mass of Oxygen} = \frac{\text{Molar Mass}}{Avogadro's\ Number}$$

$$m = \frac{M}{N_A} = \frac{0.032}{6.02 \times 10^{23}}$$

$$m = 5.31 \times 10^{-26}$$

$$\because v_{r.m.s} = \sqrt{\frac{3KT}{m}}$$

$$v_{r.m.s} = \sqrt{\frac{3(1.38 \times 10^{-23})(800)}{(5.31 \times 10^{-26})}}$$

$$v_{r.m.s} = \sqrt{\frac{3 \times 1.38 \times 800}{5.31} \times 10^{-23+26}}$$

$$v_{r.m.s} = \sqrt{623.728 \times 10^3}$$

$$v_{r.m.s} = 789.616\, m/s\ \ Answer$$

(viii) A 10 μF capacitor is charged to a potential difference of 220V. It is then disconnected from the battery. Its plates are then connected in parallels to another capacitor and it is found that, the potential difference falls to 100V. What is the capacitance of the second capacitance of the second capacitor?

DATA:

$C_1 = 10\ \mu F$

$V_1 = 200\ V$

$V_2 = 100\ V$

$C_2 = ?$

SOLUTION:

Charge on 1st capacitor before connecting

$$\because Q_1 = C_1 V_1$$

$$Q_1 = (10)(200)$$

$$Q_1 = 2000\ C$$

Charge on 1st capacitor after connecting

$$Q_1' = C_1 V_1$$

$$Q_1' = (10)(100)$$

$$Q_1' = 1000\ C$$

Charge Store on Q_2 = Charge transfer from Q_1 to Q_2

$$Q_2 = Q_1 - Q_1'$$

$$Q_2 = 2000 - 1000$$

$$Q_2 = 1000$$

Now

$$Q_2 = C_2 V_2$$

$$C_2 = \frac{Q_2}{V_2}$$

$$C_2 = \frac{1000}{100}$$

$$C_2 = 10\ \mu F\ \ Answer$$

(ix) In a TV picture tube, an electron is accelerated by a potential difference of 12000v. Determine the De-Broglie's wavelength given that $h = 6.63 \times 10^{-34}$ JS, $e = 1.6 \times 10^{-19}$ coulombs $m_e = 9.11 \times 10^{-31}$ kg.

DATA:

$V = 12000\ volts$

$h = 6.63 \times 10^{-34} JS$
$e = 1.6 \times 10^{-19}$coulombs
$m_e = 9.11 \times 10^{-31}$kg.
$\lambda = ?$
SOLUTION:

$$\because V = \sqrt{\frac{2Ve}{m_e}}$$
$$V = \sqrt{\frac{2(12000)(1.6 \times 10^{-19})}{9.11 \times 10^{-31}}}$$
$$V = \sqrt{\frac{3.84 \times 10^{-15}}{9.11 \times 10^{-31}}}$$
$$V = \sqrt{4.215 \times 10^{15}}$$
$$V = 6.492 \times 10^{7} m/s$$

$$\because \lambda = \frac{h}{m_e V}$$
$$\lambda = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(6.492 \times 10^{7})}$$
$$\lambda = \frac{6.63}{59.142} \times 10^{-34+31-7}$$
$$\lambda = \frac{6.63}{59.142} \times 10^{-34+31-7}$$
$$\lambda = 1.121 \times 10^{-11}$$
$$\lambda = \frac{1.121 \times 10^{-11}}{1 \times 10^{-10}}$$
$$\lambda = 0.112 A^{\circ} \quad Answer$$

(x) Determine the longest and the shortest wavelength photons emitted in the Lyman series ($R_H = 1.097 \times 10^{7}/m$).
DATA:
$\lambda_{Shortest} = ?$
$\lambda_{Longest} = ?$
For Shortest wavelength $n_i = \infty$
For longest wavelength $n_i' = 1$
$n_f = 1$
$R_H = 1.097 \times 10^{7}/m$
SOLUTION:
FOR SHORTEST WAVELENGTH

$$\because \frac{1}{\lambda_{Shortest}} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$
$$\frac{1}{\lambda_{Shortest}} = (1.097 \times 10^{7})\left(\frac{1}{(1)^2} - \frac{1}{(\infty)^2}\right)$$
$$\frac{1}{\lambda_{Shortest}} = 1.097 \times 10^{7}\left(\frac{1}{1} - 0\right)$$
$$\frac{1}{\lambda_{Shortest}} = 1.097 \times 10^{7}\left(\frac{1}{1}\right)$$
$$\frac{1}{\lambda_{Shortest}} = 1.097 \times 10^{7}$$
$$\lambda_{Shortest} = \frac{1}{1.097} \times 10^{-7}$$
$$\lambda_{Shortest} = 9.115 \times 10^{-8} Meter$$

FOR LONGEST WAVELENGTH

$$\frac{1}{\lambda_{longest}} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i'^2}\right)$$

$$\frac{1}{\lambda_{longest}} = (1.097 \times 10^7)\left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.097 \times 10^7\left(\frac{1}{1} - \frac{1}{4}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.097 \times 10^7\left(\frac{4-1}{4}\right)$$

$$\frac{1}{\lambda_{longest}} = 1.097 \times 10^7\left(\frac{3}{4}\right)$$

$$\frac{1}{\lambda_{longest}} = 8.227 \times 10^6$$

$$\lambda_{longest} = \frac{1}{8.227} \times 10^{-6}$$

$$\lambda_{longest} = 1.215 \times 10^{-7}\,Meter \quad Answer$$

PHYSICS 2013

(iv) An Alternating current generator operates at 79Hz. The area of the coil is 500cm². Calculate the number of turns in the coil when a magnetic field of induction 0.06 web/m². produces a maximum potential differences of 149 volts.

DATA:

$Angular\ Speed = \omega = 79Hz$

$A = 500cm^2 = \frac{500}{100 \times 100} = 0.05m^2$

$B = 0.06\, web/m^2$

$E_{max} = 149\, Volts$

$N? =$

SOLUTION:

$$\because E_{max} = BNA\omega$$

$$N = \frac{E_{max}}{BA\omega}$$

$$N = \frac{149}{(0.06)(0.05)(79)}$$

$$N = \frac{149}{0.237}$$

$$N = 628\, turns \quad Answer$$

(v) A hydrogen atom in the ground state get excited by absorbing a photon of 12.15 eV. Find the quantum number of this state.

DATA:

$E_1 = -13.6\, eV$

$E_2 = 12.15eV$

$n = ?$

SOLUTION:

$$\because E_n = E_1 + E_2$$
$$E_n = 13.6 + 12.15$$
$$E_n = 1-.45\ eV$$

Energy of electron in any state is given by

$$E_n = \frac{1}{n^2}(-13.6)eV$$
$$-1.45\ eV = \frac{1}{n^2}(-13.6)eV$$
$$n^2 = \frac{13.6}{1.45}$$
$$n^2 = 9.38$$
$$n = 3.06$$

Quantum number of the excited state is 3

(vi) Pair annihilation occurred due to a head-on-collision of an electron an a positron having the same kinetic energy, producing pair of photons ach hav'ng of 2.5 MeV. What were their kinetic energies before collision? Given m_oc^2=0.511 MeV

DATA:

Energy of each photon $= hv = 2.5\ Mev.$

$m_oc^2 = 0.511$ MeV

$K.E. = ?$

$E =$

SOLUTION:

For pair annihilation

$$2hv = 2m_oc^2 + K.E_{-e} + K.E_{+e}$$
$$2(2.5) = 2(0.511) + K.E_{-e} + K.E_{+e}$$
$$5 = 1.022 + K.E_{-e} + K.E_{+e}$$
$$5 - 1.022 = K.E_{-e} + K.E_{+e}$$
$$3.978 = K.E_{-e} + K.E_{+e}$$

But in this case energy of $K.E_{-e} = K.E_{+e}$

$$3.978 = 2K.E$$
$$K.E. = 1.989\ MeV$$

(vii) The number of atoms per gram of $_{88}Ra^{226}$ is 2.666×10^{21} and it decay with a half life of 1622 years; find the Activity and Decay constant of the sample.

(See Question No. 2 (viii), 2016)

(x) The difference of temperature between a hot and a cold body is 120°C. If the heat engine is 30% efficient, find the temperature of the hot and the cold body.

DATA:

$\Delta T = 120°C = 120 + 273 = 393K$

$E = 30\%$

$T_1 = ?$

$T_2 = ?$

SOLUTION:

$$\because E = 1 - \frac{T_2}{T_1}$$
$$E = \frac{T_1 - T_2}{T_1}$$

$$T_1 = \frac{\Delta T}{E}$$

$$T_1 = \frac{393}{\frac{30}{100}}$$

$$T_1 = \frac{393}{0.3}$$

$$T_1 = 1310K$$

$$\Delta T = T_1 - T_2$$

$$T_2 = T_1 - \Delta T$$

$$T_2 = 1310 - 393$$

$$T_2 = 917\,K\ Answer$$

(xi) The surface charge density on a vertical metal plate is $25 \times 10^{-6}\ c/m^2$. Find the force experienced by a charge of $2 \times 10^{-10}\ c$ placed in front close to the sheet. $[\varepsilon_o = 8.85 \times 10^{-12}\ C\ /\ Nm^2]$

DATA:

$\sigma = 25 \times 10^{-6}\ c/m^2$

$q = 2 \times 10^{-10}\ c$

$\varepsilon_o = 8.85 \times 10^{-12}\ C\ /\ Nm^2$

$F = ?$

SOLUTION:

$$\because F = Eq - - - (1)$$

$$E = \frac{\sigma}{2\varepsilon_o}$$

$$E = \frac{(25 \times 10^{-6})}{2(8.85 \times 10^{-12})}$$

$$E = \frac{25}{17.7} \times 10^{-6+12}$$

$$E = 1.412 \times 10^6$$

Substitute the value of "E" in equation (1)

$$F = Eq$$

$$F = (1.412 \times 10^6)(2 \times 10^{-10})$$

$$F = 2.824 \times 10^{-4} N\ \ Answer$$

(xii) Two resistors of 5Ω and 2Ω are connected in parallel with a 9V battery. Calculate the current and power dissipated in each resistance.

DATA:

$R_1 = 5\Omega$

$R_2 = 2\Omega$

$V_{ab} = 9\ Volts$

$I_1 = ?$

$I_2 = ?$

$P_1 = ?$

$P_2 = ?$

SOLUTION:

Current through 1st resistors

$$\because V_{ab} = IR$$
$$V_{ab} = I_1 R_1$$
$$I_1 = \frac{V_{ab}}{R_1}$$
$$I_1 = \frac{9}{5}$$
$$I_1 = 1.8\ Ampere$$

Current through 2nd resistors

$$V_{ab} = I_1 R_2$$
$$I_2 = \frac{V_{ab}}{R_2}$$
$$I_1 = \frac{9}{2}$$
$$I_1 = 4.5\ Ampere$$

Power dissipated through 1st resistors

$$\because P_1 = V_{ab} I$$
$$P_1 = (9)(1.8)$$
$$P_1 = 16.2\ watt$$

Power dissipated through 2nd resistors

$$P_2 = V_{ab} I$$
$$P_2 = (9)(4.5)$$
$$P_2 = 40.5\ watt \quad Answer$$

(xv) A voltmeter measuring upto 200 volts has a total resistance of 20,000 ohms. What additional series resistance must be connected to it to increase its range to 600 volts?

DATA:

$V_1 = 200\ volts$

$R_v = 20,000\Omega$

$V_2 = 600\ volts$

$R_x =$

SOLUTIO

$$\because V_1 = IR_v$$
$$I = \frac{V_1}{R_x}$$
$$I = \frac{200}{20000} = 0.1\ Ampere$$
$$\because R_x = \frac{V_2}{I} - R_v$$
$$R_x = \frac{600}{0.1} - 20000$$
$$R_x = 60000 - 20000$$
$$R_x = 40000\Omega \quad Answer$$